Nov 10, 2024 (Due: 08:00 Nov 17, 2025)

1. Given a skew-Hermitian matrix A. Describe how to compute all eigenvalues and eigenvectors of A?

2. Let A and E be Hermitian matrices with $AE = EA$. Try to give an upper bound on

$$\|\exp(A+E) - \exp(A)\|_2.$$

Make sure your upper bound tends to zero when $\|E\|_2 \to 0$.

3. Implement the scaling-and-squaring algorithm (combined with truncated Taylor series) for computing the matrix exponential. Test the accuracy of your algorithm by a few diagonalizable matrices with known spectral decomposition.

(optional) Implement the Schur–Parlett algorithm and compare the accuracy.

(H, optional) Replace truncated Taylor series by Padé approximants in the scaling-and-squaring algorithm and compare the accuracy. The formulae for Padé approximants of the exponential function can be found in the paper "Nineteen dubious ways to compute the exponential of a matrix" by C. Moler and C. F. Van Loan (available on eLearning).

4. Consider the iterative scheme

$$X_{k+1} = \frac{X_k + X_k^{-1}A}{2}, \qquad X_0 = I,$$

where A is a given Hermitian positive definite matrix. Show that, in theory, $\lim_{k\to\infty} X_k = A^{1/2}$.

Try this iterative scheme on some randomly generated examples. Do the experimental results agree with the theoretical prediction?

5. Read the paper "From Random Polygon to Ellipse: An Eigenanalysis" by A. N. Elmachtoub and C. F. Van Loan (available on eLearning). Reproduce the experiments in this paper.

6. (optional) In the lecture we have discussed how to solve Sylvester matrix equation $AX - XB = C$ using the Bartels–Stewart algorithm (through Schur decompositions). What happens the matrices are real, and if only *real* Schur decompositions are permitted?

7. (optional) Suppose that the Lyaponov equation $XA + A^*X = -Q$ has a unique solution. By slightly perturbed the inputs, we obtain

$$(X + \Delta X)(A + \Delta A) + (A + \Delta A)^*(X + \Delta X) = -Q - \Delta Q.$$

Define the notation $\mathrm{Sep}(\cdot,\cdot)$ as

$$\mathrm{Sep}(U, V) = \min_{\|Y\|_2=1} \|UY - YV\|_2.$$

Show that if $2\|\Delta A\|_2 < \mathrm{Sep}(A^*, -A)$, then

$$\frac{\|\Delta X\|_2}{\|X\|_2} \leq \frac{2\|A\|_2}{\mathrm{Sep}(A^*, -A) - 2\|\Delta A\|_2} \left(\frac{\|\Delta A\|_2}{\|A\|_2} + \frac{\|\Delta Q\|_2}{\|Q\|_2} \right).$$